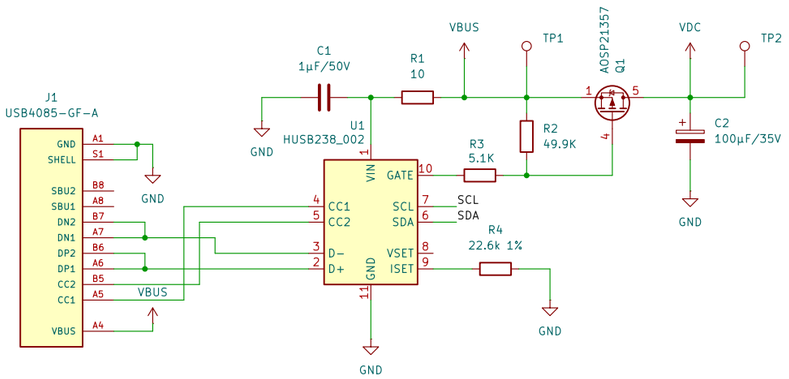
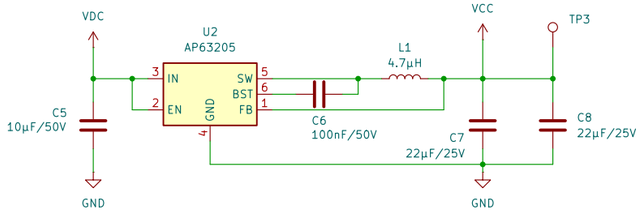
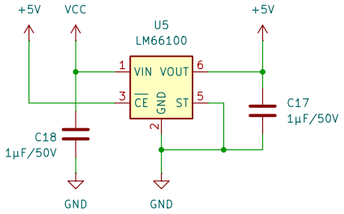
USB-C sink controller, configured to negotiate a 20V contract at 3A



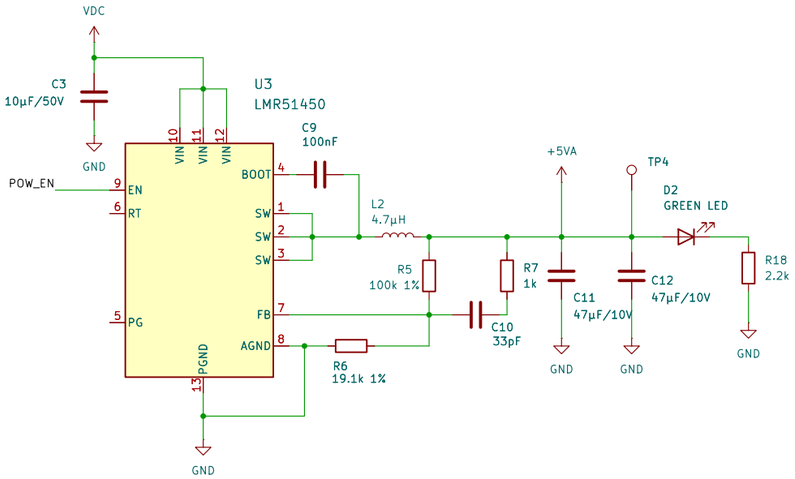
5V @2A step-down converter to power MatrixPortal and ICs

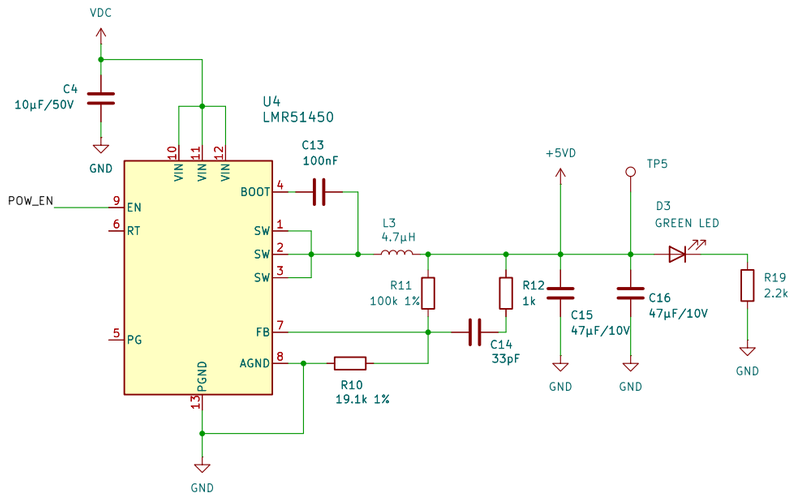


Ideal diode wired for reverse current blocking

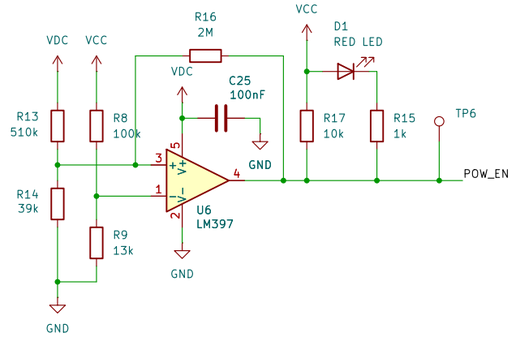


Pair of step-down converters to output 5V @ 5A on each of the power rails

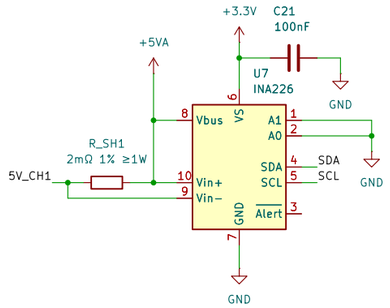


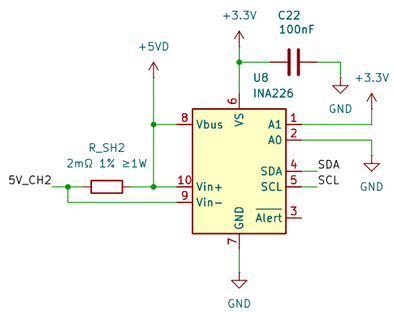


Voltage comparator to determine whether the USB-C contract is good enough to enable the 5V power rails

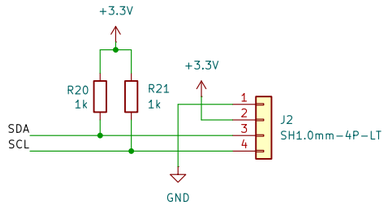


Current sensors connected to the Matrix Portal via I2C

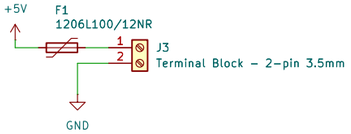




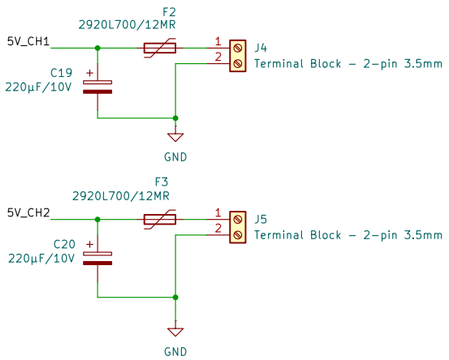
Connects to the StemmaQT port on the MatrixPortal



Connects to the screw terminals on the Matrix Portal for 5V supply



Connects to the dedicated power connector on each matrix panel



MRC		
Sheet: /		
File: powerboard.kicad_sch		
Title: MatrixChain Power Board		
Size: A3	Date: 2026-11-08	**Rev: Rev A.**
KiCad E.D.A. 10.0.8		Id: 1/1